What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Carbapenems

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Learning Objectives (3)

After completing this brick, you will be able to:

1. Describe the mechanism of action of carbapenems.
2. Explain the purpose of administering imipenem with cilastatin and the potential adverse effects of high levels of imipenem.
3. Recognize the most common adverse drug reactions and drug interactions associated with carbapenems.

CASE CONNECTION

SG is a 62-year-old female admitted to the hospital with fever, flank pain, and dysuria for the past three days. She has a history of type 2 diabetes mellitus and recurrent urinary tract infections (UTIs), for which she has been treated with multiple courses of antibiotics in the past year. On examination, she is febrile (102.1°F), tachycardic, and has tenderness over her right costovertebral angle. Her urinalysis reveals pyuria, bacteriuria, and positive nitrites, and she has a history of extended-spectrum beta-lactamase (ESBL)-producing organisms. As you are combing through her chart to determine which empiric antibiotic to start you also note that she has a severe (anaphylactic) allergy to penicillin.

As a medical student on your internal medicine clerkship, your attending asks you which antibiotic should be started empirically? SG is also concerned because of her allergy, she states "don't give me any penicillin!" How would you justify your empiric antibiotic choice to SG?

GO TO CONCLUSION ↓

How do carbapenems and monobactams work? (LO#1)

Like the other β-lactam antibiotics we have discussed, carbapenems kill bacteria by altering the cell wall. Notice from the image below (Figure 1), that the β-lactam ring remains in the hallmark structure in the carbapenem class. This class of antibiotics has a broader spectrum of activity than most other β-lactam antibiotics largely due to their greater resistance to β-lactamase–mediated hydrolysis. The most commonly available carbapenems include meropenem, /cilastatin. Each of these are summarized in Table 1 below.

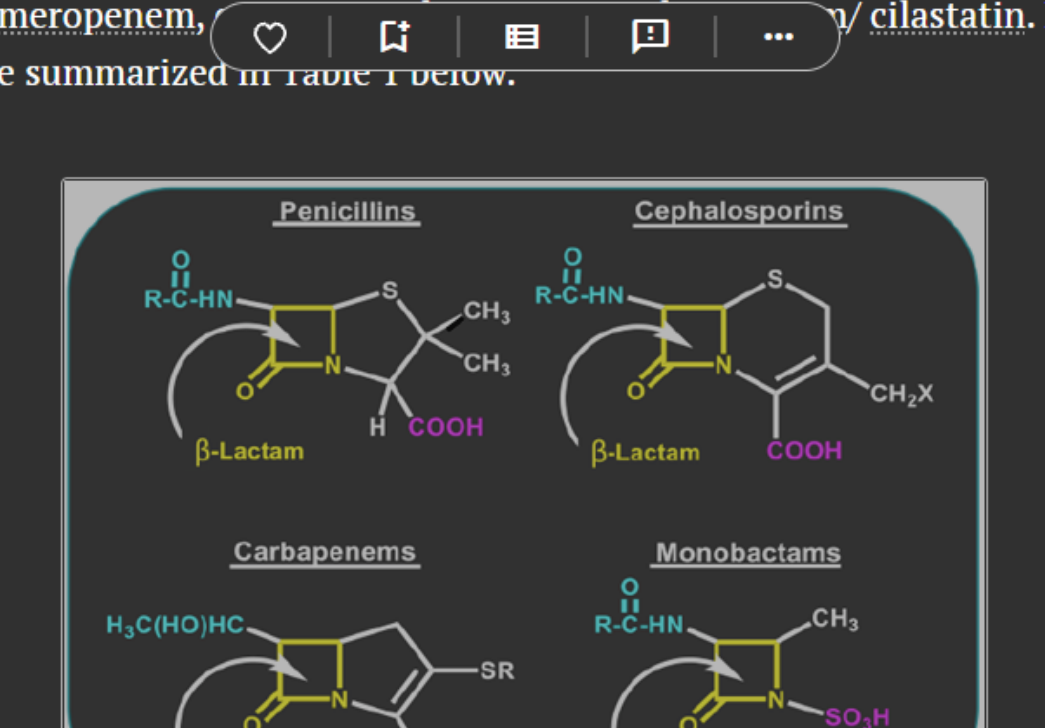

Figure 1

How do β-lactam antibiotics kill bacteria?

The core structure carbapenems, just like penicillins and the other β-lactam antibiotics, is the β-lactam ring, which binds to and irreversibly (noncompetitively) inhibits the transpeptidase (or penicillin-binding protein [PBP]) enzyme, the enzyme that cross-links the linear peptidoglycan chain components of the cell wall (Figure 2). Cell wall synthesis is halted, and the bacteria die.

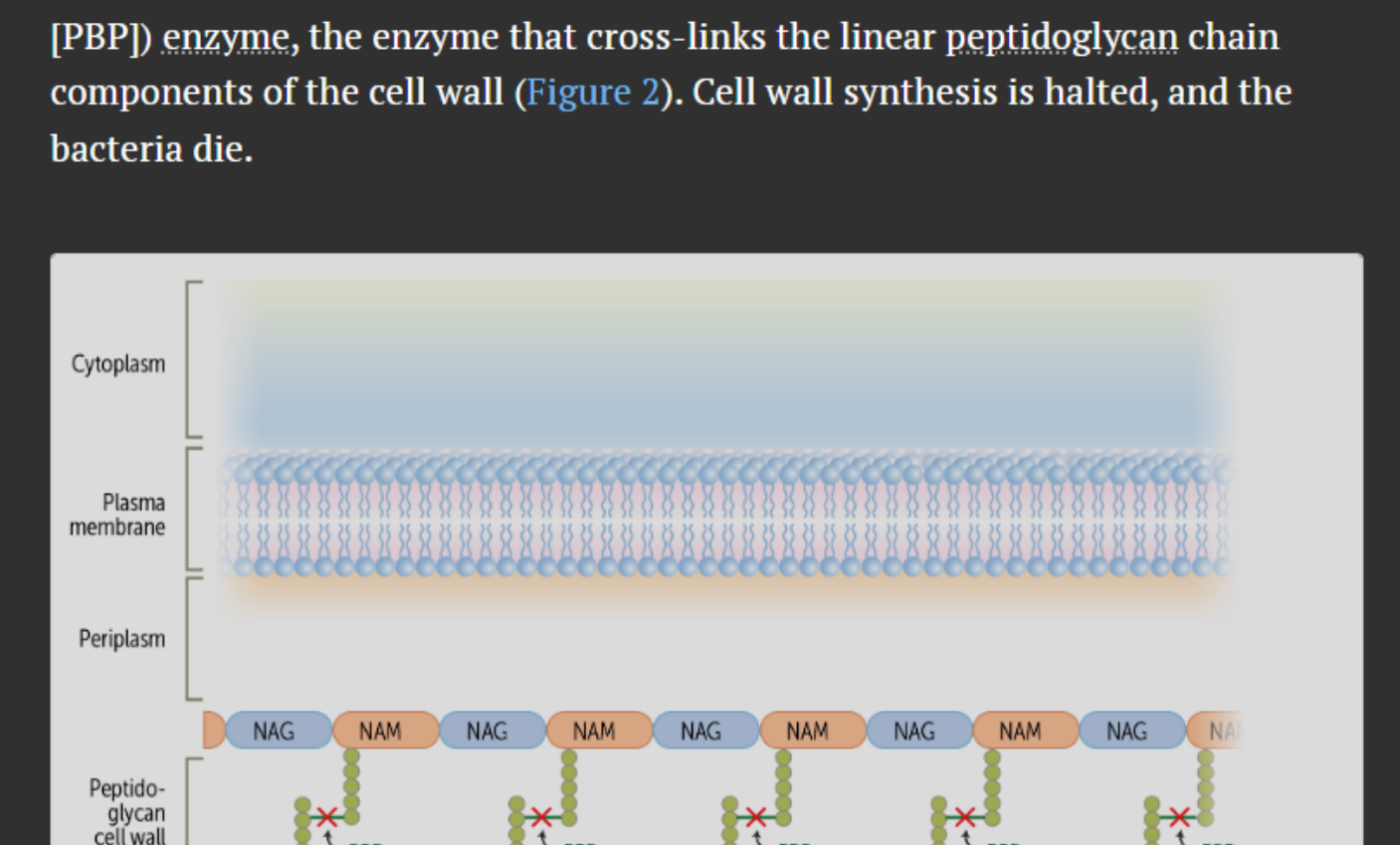

Figure 2

The β-lactam ring acts as a structural analogue for D-Ala-D-Ala, found in the peptidoglycan, and bi ocking peptidoglycan **cross-linking** in the activate autolytic enzymes (autolysins) that cause lesions in the bacterial cell wall, leading to osmotic death or killing of bacteria from the bactericidal effect of the antibiotic. Carbapenem action is dependent on the *rapid growth of bacteria*, as cell walls must be built for bacterial growth. If cell wall synthesis is inhibited by β-lactams and glycopeptides, bacterial growth is halted. These antibiotics are bactericidal, meaning they kill bacteria, but they can only do so during rapid growth when the bacteria are actively multiplying.

The carbapenems are only available in intravenous formulations. This means that they have a high bioavailability and they readily penetrate into most tissues and body fluids. In particular, the urinary tract and cerebrospinal fluid. Part of the reason they readily accumulate in the urinary tract is because they are eliminated renally. While this can be good for treating complicated urinary tract infections, it means that doses must be adjusted for patients with renal impairment. This means doses need to be lowered in patients with poor renal function because they do not readily clear the medication as quickly as patients with normal kidney function. Further, if doses are not lowered in patients with poor renal function it can make them much more susceptible to experience side effects, which will be discussed in a section below. The pharmacokinetics of imipenem/ cilastatin differ slightly from the other agents in this class, however, that will be discussed in the next section.

CLINICAL CORRELATION

Carbapenems are another class of broad-spectrum antibiotics, possessing incredible coverage over gram-positive cocci, gram-negative rods, and anaerobes. Some carbapenems ffective against *Pseudomonas*. Because are typically considered "last-resort" drugs for patients with serious, life-threatening illnesses. Or, another way to think of it is that we reserve carbapenems for extended spectrum beta-lactamase (ESBLs) producing organisms. But, what if the bacteria are resistant to carbapenems?

There is a newer antibiotic, or rather a new combination, that includes meropenem and the beta-lactamase inhibitor vaborbactam (i.e., meropenem/ vaborbactam). This combination, just like you have learned with some of the penicillin combinations (e.g., amoxicillin/ clavulanate and piperacillin/ tazobactam), extends the coverage of the beta-lactam and can prevent the beta-lactamase producing organisms from rendering the antibiotic ineffective. This drug is expensive and specifically reserved for very resistant organisms, however, you may see it in your future practice.

How do monobactams work?

Monobactams are β-lactams that contain only a fused β-lactam ring (Figure 1), not a thiazolidine or dihydrothiazidine ring. Currently, aztreonam is the only member of this class in therapeutic use.

Monobactams have a similar mechanism of action to the other β-lactams. They inhibit bacterial cell wall synthesis by binding to PBPs, which prevents the final step of peptidoglycan synthesis in bacterial cell walls.

Because of the unique monobactam structure, aztreonam is primarily reserved for patients with severe penicillin/ β-lactam allergies. The monobactam structure makes cross-reactivity with other β-lactams unlikely, so it tends to be reserved for those situations. Unlike the carbapenems, that should still be avoided in severe/ an w rates of cross reactivity.

Just like the carbapenems, aztreonam is available in intravenous formulation. It has high bioavailability and penetrates the blood brain barrier to accumulate in the cerebrospinal fluid. Also, just like the carbapenems, it is eliminated, and accumulates, in the urine. So, dosage adjustments are needed for patients with renal impairment.

CLINICAL CORRELATION

Aztreonam is also available as an inhaled solution, but this is reserved for patients with cystic fibrosis. You will learn more about this in the respiratory course, however, these patients are prone to multi-drug resistant respiratory infections and inhaled aztreonam can help.

Why do imipenem and cilastatin have to be given together? (LO#2)

Imipenem, like the other carbapenems, is an intravenous medication. When administered, the drug is hydrolyzed rapidly by a dipeptidase found in the brush border of the proximal tubule of the nephron. If imipenem is given without cilastatin it essentially becomes ineffective because the hydrolyzation greatly reduces the half-life. To combat this, imipenem is formulated in combination with cilastatin (imipenem/cilastatin), to inhibit the degradation of imipenem by a renal tubular dipeptidase and extend its elimination half-life.

This combination of rmacokinetic standpoint, is very similar to the other carbapenems. It has good bioavailability and penetrates many different body tissues and fluids, including cerebrospinal fluid. It is primarily eliminated in the urine, and, of course, requires dosage adjustment for patients with renal impairment.

CLINICAL CORRELATION

Just like we discussed above with meropenem/ vaborbactam. Now, imipenem is available as a coformulation with the β-lactamase inhibitor relebactam (imipenem/cilastatin/relebactam), which extends its spectrum against organisms producing carbapenemases.

Why is cilastatin combined with imipenem?

What are the side effects of carbapenems and monobactams? (LO#3)

Common side effects associated with carbapenems and monobactams include diarrhea, nausea, vomiting, and they can increase liver function tests. Although rare there are a few more serious side effects to be aware of:

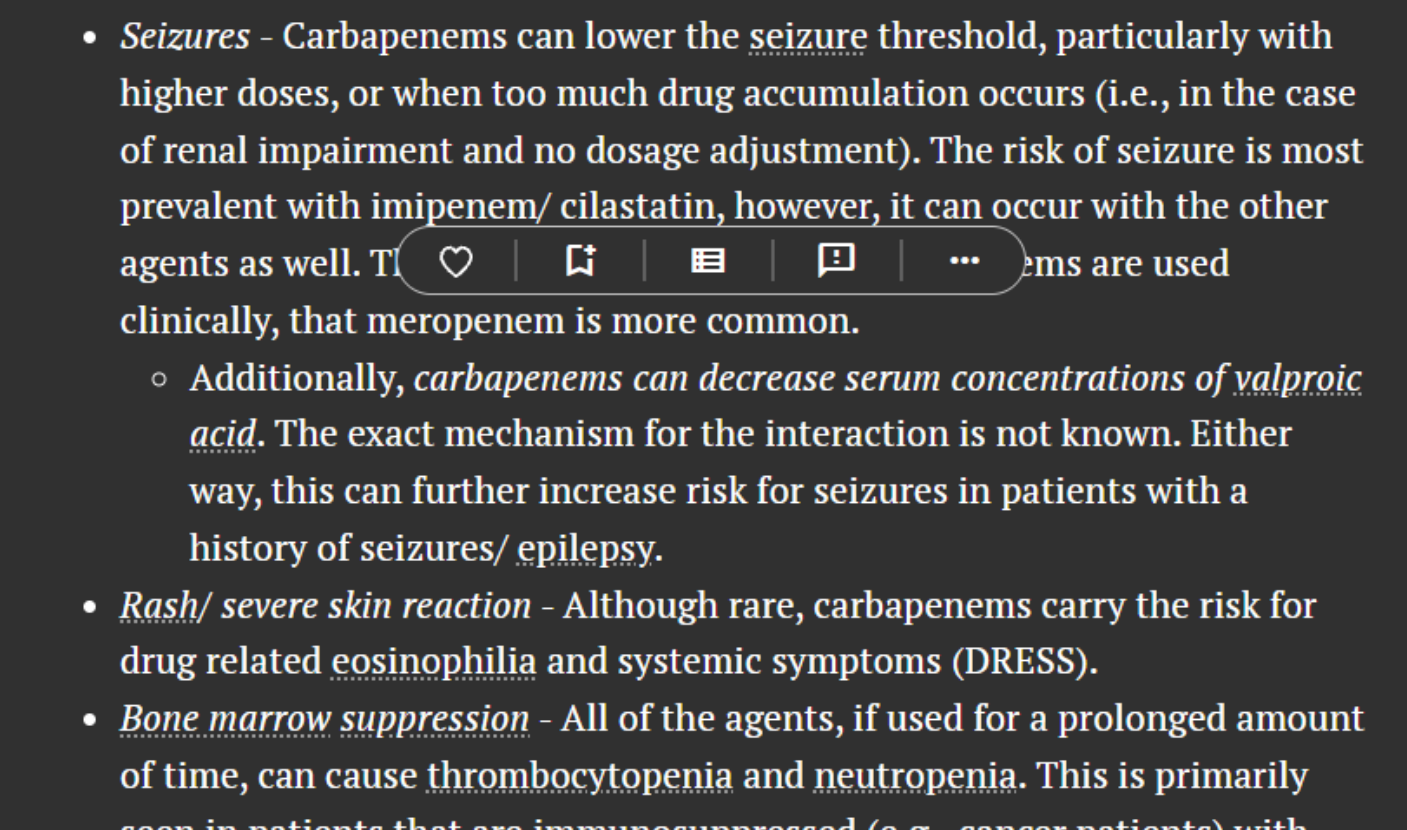
Common Adverse Reactions of: Carbapenems & Monobactams
Bone Marrow Suppression
Seizures
Rash

Figure 3

- *Seizures* - Carbapenems can lower the seizure threshold, particularly with higher doses, or when too much drug accumulation occurs (i.e., in the case of renal impairment and no dosage adjustment). The risk of seizure is most prevalent with imipenem/ cilastatin, however, it can occur with the other agents as well. T ems are used clinically, that meropenem is more common.
    - Additionally, *carbapenems can decrease serum concentrations of valproic acid*. The exact mechanism for the interaction is not known. Either way, this can further increase risk for seizures in patients with a history of seizures/ epilepsy.
- *Rash/ severe skin reaction* - Although rare, carbapenems carry the risk for drug related eosinophilia and systemic symptoms (DRESS).
- *Bone marrow suppression* - All of the agents, if used for a prolonged amount of time, can cause thrombocytopenia and neutropenia. This is primarily seen in patients that are immunosuppressed (e.g., cancer patients) with multi-drug resistant infections, that require carbapenems for a long period of time.

Which carbapenem has the highest risk of causing seizures?

BACK TO INTRODUCTION ↑

CASE CONNECTION

Returning to SG.

She has been admitted to the hospital with a UTI, and, given her history of recurrent UTI, frequent antibiotic use, and past infection with ESBLs, it is likely that this is another ESBL organism. Carbapenems, like meropenem, are the drug of choice for ESBLs, even empirically they are warranted in SG's case, however, she has a severe penicillin allergy. Despite relatively low cross reactivity rates with the carbapenems, they should still be avoided in patients with with severe allergy. This would be an great opportunity to utilize aztreonam, the monobactam. It retains coverage against many ESBLs, and given it's structure as a monobactam it does not have any cross reactivity in those with penicillin allergy.

As you empathize with SG about her allergy, and not wanting to make her have an allergic reaction, you explain that the chemical structure of aztreonam is different from the penicillins and it does not have any cross reactivity with her allergy.

Summary

- Carbapenems and monobactams are β-lactam antibiotics that inhibit bacterial cell wall synthesis by binding to penicillin-binding proteins (PBPs).
- Carbapenems have a broad spectrum of activity and are often reserved for serious, life-threatening infections due to their resistance to β-lactamase–mediated hydrolysis.
- Monobactams, such as aztreonam, are primarily used for patients with severe penicillin allergies due to their unique structure.
- Imipenem is com degradation by renal dipeptidase, extending its half-life and effectiveness.
- Common side effects of carbapenems and monobactams include gastrointestinal symptoms and increased liver function tests, with more serious risks like seizures and bone marrow suppression.
- Dosage adjustments for carbapenems and monobactams are necessary in patients with renal impairment to prevent drug accumulation and associated side effects.

Review Questions

1. Which carbapenem has the highest risk of causing seizures?

A. ● Imipenem/ cilastatin
B. ○ Meropenem
C. ○ Ertapenem
D. ○ Doripenem
E. ○ Aztreonam

Hide Explanation

The correct answer is imipenem/cilastatin (A). Imipenem/cilastatin is known to have the highest risk of causing seizures, especially at higher doses or in patients with renal impairment. Meropenem (B) is used more commonly due to its lower seizure risk. Ertapenem (C) a mpared to imipenem/cilastatin. Aztreonam (E) is a monobactam and not a carbapenem, so it is not relevant to this question.

2. Which antibiotic is primarily used for patients with severe penicillin allergies?

A. ● Aztreonam
B. ○ Meropenem
C. ○ Ertapenem
D. ○ Doripenem
E. ○ Imipenem/Cilastatin

Hide Explanation

The correct answer is Aztreonam (A). Due to its unique monobactam structure, aztreonam is primarily used for patients with severe penicillin allergies. Meropenem (B), Ertapenem (C), Doripenem (D), and Imipenem/Cilastatin (E) are carbapenems, not monobactams, and are not primarily used for patients with severe penicillin allergies.

3. Why is imipenem administered with cilastatin?

A. ○ To increase bioavailability
B. ● To prevent rapid degradation by renal dipeptidase
C. ○ To reduce side effects
D. ○ To enhance penetration into tissues

Hide Explanation

The correct answer is to prevent rapid degradation by renal dipeptidase (B). Imipenem is hydrolyzed rapidly by a dipeptidase found in the brush border of the proximal tubule of the nephron. To extend its half-life and effectiveness, it is combined with cilastatin, which inhibits this degradation. It is not because of increasing bioavailability (A), reducing side effects (C), or enhancing penetration into tissues (D).

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

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01 ☑ ~~Describe the mechanism of action of carbapenems.~~
02 ☑ ~~Explain the purpose of administering imipenem with cilastatin and the potential adverse effects of high levels of imipenem.~~
03 ☑ ~~Recognize the most common adverse drug reactions and drug interactions associated with carbapenems.~~

Objective Confidence: 3/3